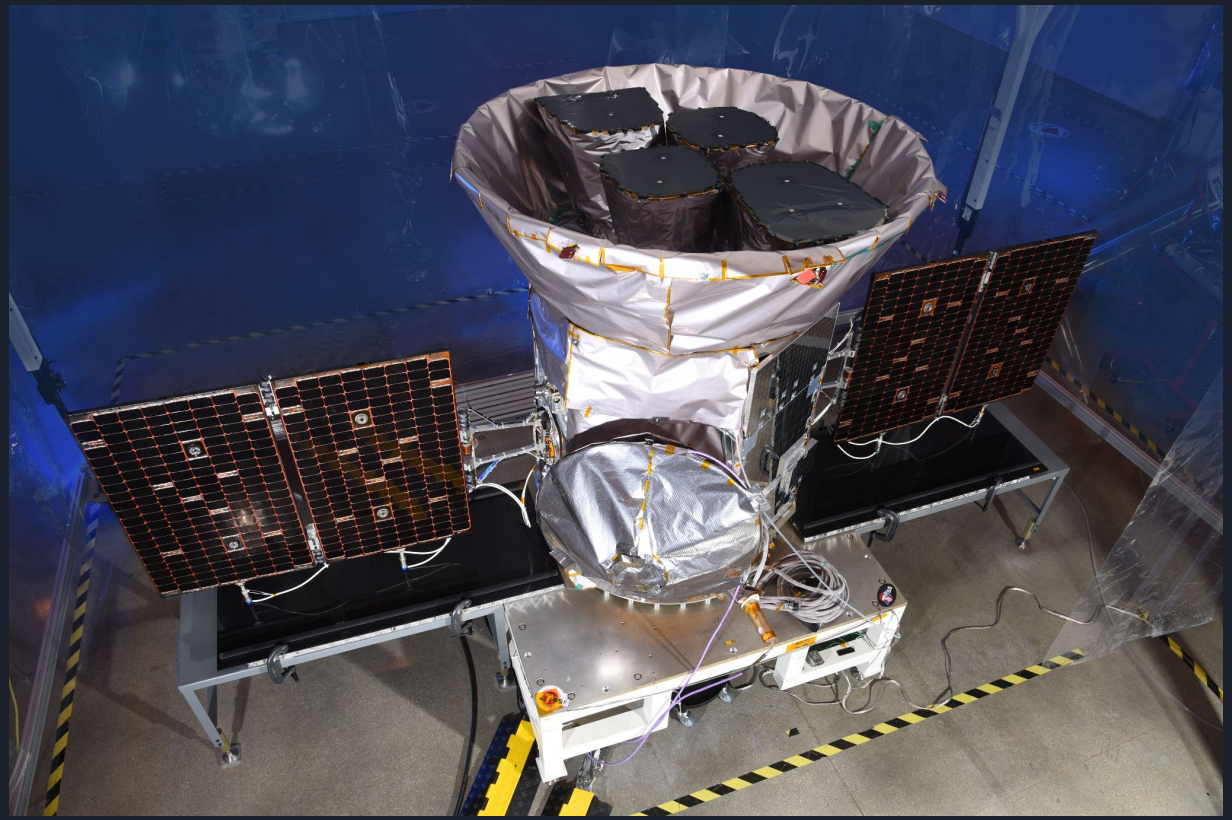
Mapping the Universe

Jonah Lutchmedial

TESS



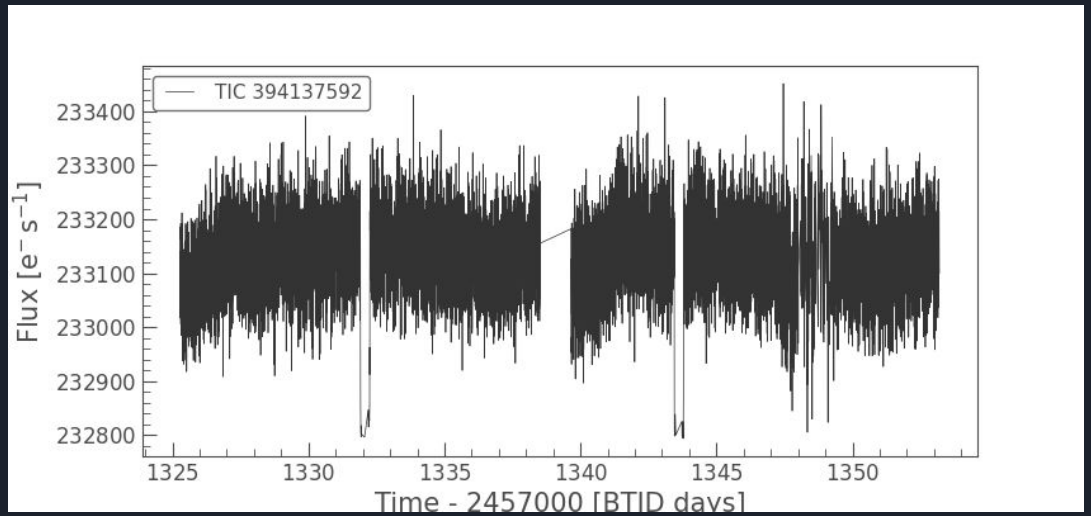
My Inspiration

IDENTIFYING EXOPLANETS WITH DEEP LEARNING: A FIVE PLANET RESONANT CHAIN AROUND KEPLER-80 AND AN EIGHTH PLANET AROUND KEPLER-90

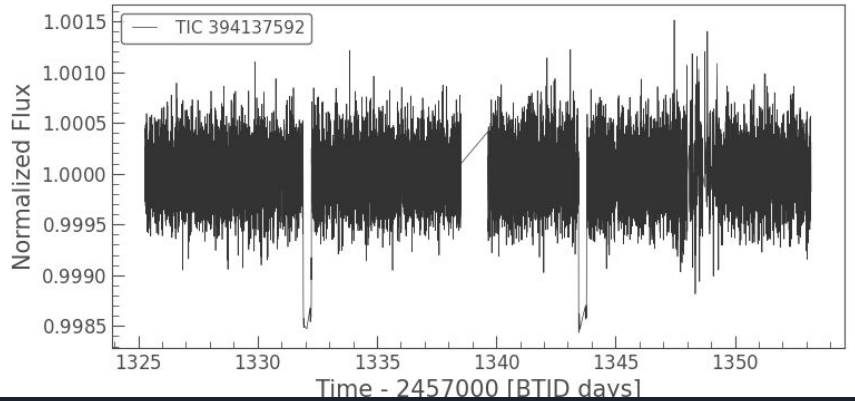
CHRISTOPHER J. SHALLUE[†,1] & ANDREW VANDERBURG[⋆,2,3]

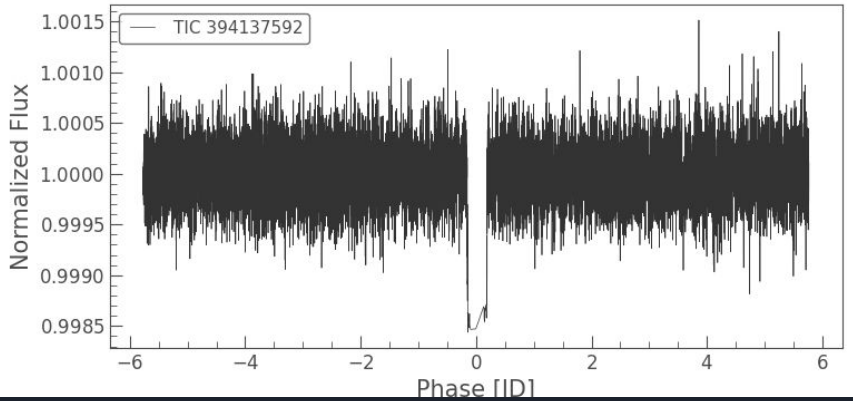
TESS + Kepler Missions

- NASA Mission taking 27 days of data on different areas of the sky
- Isolate stars into 11x11 px squares
- Track brightness over time



Normalize -> (BLS) Folding





What was my plan?

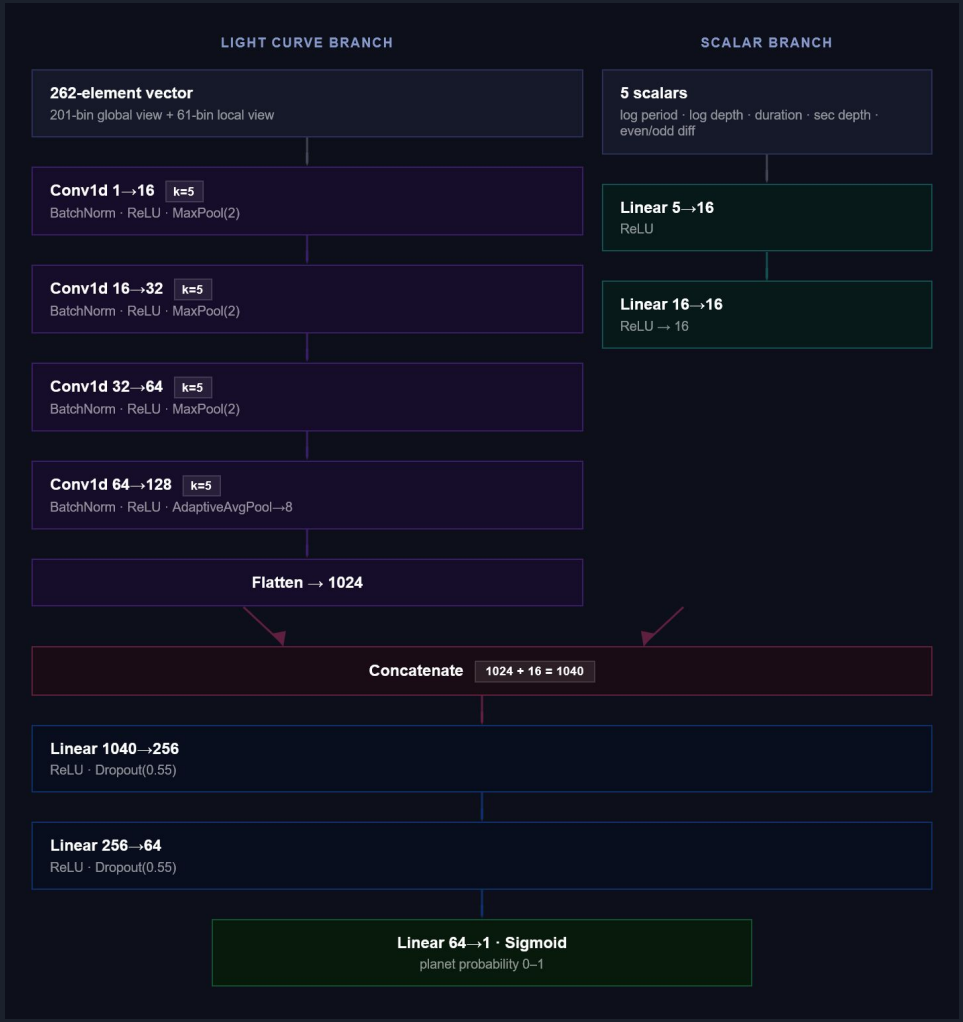
1. Global view (201 values)
2. Local view (61 values)
3. Secondary Transit (1 Value)
4. Even/odd depth difference (1 value)

Plus the **3 scalar features** fed into the separate branch:

- `log(period)` in days
- `log(depth)` in ppm
- `duration` in hours

→ 2 Branch,
4 Layer CNN

→ Is this a planet



ReLU: Applies the learning via eliminating negative features

MaxPool:
Save only the best features

AvgPool:
Average the vectors to 8 elements

```
Device: cuda
GPU: NVIDIA GeForce RTX 5070

Loading TESS cache from 'lc_cache'...
Loaded 4,301 samples  |  Planets: 1,404  FPs: 2,897
Train: 3,106  |  Val: 549  |  Test: 646

  Class weights: pos_weight=2.063  (neg=2092 pos=1014)
Training  |  3106 train  549 val  |  epochs=200  batch=32  lr=0.001

  Epoch   1/200  loss=0.8932  prec=0.518  rec=0.637  F1=0.571  *
  Epoch   5/200  loss=0.8642  prec=0.530  rec=0.642  F1=0.581
  Epoch  10/200  loss=0.8467  prec=0.581  rec=0.659  F1=0.618
  Epoch  15/200  loss=0.8326  prec=0.629  rec=0.693  F1=0.660
  Epoch  20/200  loss=0.8373  prec=0.600  rec=0.704  F1=0.648
  Epoch  25/200  loss=0.8269  prec=0.624  rec=0.676  F1=0.649
  Epoch  30/200  loss=0.8215  prec=0.628  rec=0.698  F1=0.661
  Epoch  35/200  loss=0.8189  prec=0.639  rec=0.693  F1=0.665
  Epoch  40/200  loss=0.8152  prec=0.637  rec=0.687  F1=0.661
  Epoch  45/200  loss=0.8125  prec=0.640  rec=0.665  F1=0.652
  Epoch  50/200  loss=0.8105  prec=0.634  rec=0.687  F1=0.660
  Epoch  55/200  loss=0.8121  prec=0.638  rec=0.698  F1=0.667
  Epoch  60/200  loss=0.8101  prec=0.608  rec=0.721  F1=0.660
  Epoch  65/200  loss=0.8104  prec=0.638  rec=0.709  F1=0.672  *
  Epoch  70/200  loss=0.8115  prec=0.643  rec=0.704  F1=0.672  *
  Epoch  75/200  loss=0.8065  prec=0.626  rec=0.709  F1=0.665
  Epoch  80/200  loss=0.8086  prec=0.627  rec=0.715  F1=0.668
  Epoch  85/200  loss=0.8079  prec=0.620  rec=0.721  F1=0.667
  Epoch  90/200  loss=0.8080  prec=0.615  rec=0.715  F1=0.661
  Epoch  95/200  loss=0.8072  prec=0.639  rec=0.693  F1=0.665
  Epoch 100/200  loss=0.8107  prec=0.641  rec=0.687  F1=0.663
  Epoch 105/200  loss=0.8081  prec=0.641  rec=0.687  F1=0.663
  Epoch 110/200  loss=0.8087  prec=0.647  rec=0.687  F1=0.667
  Epoch 115/200  loss=0.8085  prec=0.639  rec=0.693  F1=0.665
  Early stopping at epoch 117 (no improvement for 40 epochs)

  Best val F1: 0.676
```

```
── Classification Report ──────────────────────────────
                   precision    recall  f1-score   support

  False Positive       0.86      0.83      0.85       435
Planet Candidate       0.68      0.71      0.69       211

        accuracy                           0.79       646
       macro avg       0.77      0.77      0.77       646
    weighted avg       0.80      0.79      0.80       646

── Confusion Matrix ───────────────────────────────────
                  Predicted FP  Predicted Planet
  Actual FP            363             72
  Actual Planet         61            150

Model    : exoplanet_cnn.pt
Metadata : exoplanet_cnn_meta.json
Device   : cuda
(exoplanet) PS C:\Projects\Exo-Planet-Finder>
```

```
(exoplanet) PS C:\Projects\Exo-Planet-Finder> python inference.py 149603524 --csv labeled_tess_dataset.csv

Device: cuda
Loaded model: exoplanet_cnn.pt

Found cached vector for TIC 149603524 — using cache.

── BLS Results ──────────────────────────────────
  Period          : 4.4119 days
  Duration        : 3.65 hours
  Depth           : 15062 ppm
  Sec. eclipse    : 0.000005  (high = likely EB)
  Even/odd diff   : 0.000003  (high = likely EB)
  Sectors used    : cached

── CNN Result ───────────────────────────────────
  ✅  Planet candidate
  Planet probability : 99.97%
  FP probability     : 0.03%
──────────────────────────────────────────────────
```